

Special Section:

Forcing, response, and impacts of coastal storms in a changing climate

Key Points:

- Pathways of land-sourced contaminants in the aftermath of Hurricane Florence are investigated with an ocean model with hydrology source
- The Cape Fear River plume had a westward coastal current and intermittent offshore storage modulated by wind direction and river discharge
- Beaches 100 km west saw river-sourced water quality hazards with delayed onset and long duration, from days to weeks after the storm

Correspondence to:

M. Moulton,
mmoulton@ucar.edu

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Author Contributions:

Conceptualization: Melissa Moulton, Z. George Xue, Ruoying He
Formal analysis: Melissa Moulton
Funding acquisition: John C. Warner
Methodology: Melissa Moulton, Joseph B. Zambon, John C. Warner, Daoyang Bao, Dongxiao Yin, Zafer Defne, Ruoying He, Christie Hegermiller
Project administration: John C. Warner
Software: Joseph B. Zambon
Visualization: Melissa Moulton, Zafer Defne
Writing – original draft: Melissa Moulton

Modeled Coastal-Ocean Pathways of Land-Sourced Contaminants in the Aftermath of Hurricane Florence

Melissa Moulton^{1,2} , Joseph B. Zambon³ , Z. George Xue⁴ , John C. Warner⁵ , Daoyang Bao⁴, Dongxiao Yin^{4,6} , Zafer Defne⁵ , Ruoying He³ , and Christie Hegermiller^{5,7} 

¹Climate and Global Dynamics Laboratory, U.S. National Science Foundation National Center for Atmospheric Research, Boulder, CO, USA, ²Applied Physics Laboratory, University of Washington, Seattle, WA, USA, ³Department of Marine, Earth, and Atmospheric Sciences, North Carolina State University, Raleigh, NC, USA, ⁴Department of Oceanography and Coastal Sciences, Louisiana State University, Baton Rouge, LA, USA, ⁵United States Geologic Survey, Woods Hole Coastal & Marine Science Center, Woods Hole, MA, USA, ⁶Now at Virginia Institute of Marine Science, Gloucester Point, VA, USA, ⁷Now at Department of Civil and Environmental Engineering, University of Washington, Seattle, WA, USA

Abstract Extreme precipitation during Hurricane Florence, which made landfall in North Carolina in September 2018, led to breaches of hog waste lagoons, coal ash pits, and wastewater facilities. In the weeks following the storm, freshwater discharge carried pollutants, sediment, organic matter, and debris to the coastal ocean, contributing to beach closures, algae blooms, hypoxia, and other ecosystem impacts. Here, the ocean pathways of land-sourced contaminants following Hurricane Florence are investigated using the Regional Ocean Modeling System (ROMS) with a river point source with fixed water properties from a hydrologic model (WRF-Hydro) of the Cape Fear River Basin, North Carolina's largest watershed. Patterns of contaminant transport in the coastal ocean are quantified with a finite duration tracer release based on observed flooding of agricultural and industrial facilities. A suite of synthetic events also was simulated to investigate the sensitivity of the river plume transport pathways to river discharge and wind direction. The simulated Hurricane Florence discharge event led to westward (downcoast) transport of contaminants in a coastal current, along with intermittent storage and release of material in an offshore (bulge) or eastward (upcoast) region near the river mouth, modulated by alternating upwelling and downwelling winds. The river plume patterns led to a delayed onset and long duration of contaminants affecting beaches 100 km to the west, days to weeks after the storm. Maps of the onset and duration of hypothetical water quality hazards for a range of weather conditions may provide guidance to managers on the timing of swimming/shellfishing advisories and water quality sampling.

Plain Language Summary Hurricane Florence hit the southeastern United States in September 2018, causing historic rainfall and flooding. Floodwaters carrying contaminants traveled down rivers to the ocean, causing harmful algal blooms, beach closures, and other impacts. In the Cape Fear River Basin, North Carolina's largest watershed, open-air lagoons used to store waste from industrial hog farming were breached, causing sewage to flow into the ocean in a coherent feature called a 'river plume' that moved in different patterns. Flooding impacts are increasing, but the movement of contaminants in river plumes following extreme events is not well understood. We used a numerical ocean model to simulate the Cape Fear River plume carrying a dye representing contaminants following Hurricane Florence. The plume initially traveled west hugging the coast due to Earth's rotation, alternately spreading east or offshore due to changes in the direction of winds blowing on the ocean. The river plume patterns led to a delayed onset and long duration of contaminants affecting beaches far down the coast, days to weeks after the storm. We also simulated several scenarios with different river discharge and winds. This information could help guide managers on when to perform water quality sampling and issue swimming or shellfishing advisories.

1. Introduction

Coastal ecosystems, infrastructure, and human health are increasingly vulnerable to storms and flooding as tropical cyclones are predicted to move more slowly and have larger total precipitation, and as populations and industries are becoming more concentrated in coastal regions (Intergovernmental Panel on Climate Change, 2021). In the United States, for example, recent large rainfall events associated with tropical cyclones include Matthew (2016) and Florence (2018) causing extreme precipitation in the southeast, Irene (2011), Sandy (2012), and Henri (2021) in the mid-Atlantic and northeast, Harvey (2017), Imelda (2019), Sally (2020), Fiona

Writing – review & editing:

Melissa Moulton, Joseph B. Zambon,
Z. George Xue, John C. Warner,
Daoyang Bao, Dongxiao Yin, Zafer Defne,
Ruoying He, Christie Hegermiller

(2022), and Ian (2022) in the Gulf and Caribbean, and Lane (2018) in Hawaii. Heavy rainfall associated with tropical storms often causes flooding of upstream and inland industrial and agricultural sites. For example, in North Carolina, Hurricane Floyd (1999) and Irene (2011) caused major flooding of wastewater treatment facilities, coal ash pits and other industrial waste storage sites, and concentrated animal feeding operations including hog farms that store waste in open-air lagoons (Figures 1a–1d) (Christenson et al., 2022; Miller et al., 2016; M. M. Brown et al., 2014). The downstream impacts of the resulting release of contaminants, nutrients, and debris to the coastal ocean include algae blooms, fish kills, and various human health impacts from exposure to contaminated flood waters and seafood (Mallin & Corbett, 2006; Yu et al., 2018; Paul et al., 2019; D'sa et al., 2019; Hunt & Jones, 2020; Paerl et al., 2020) (Figures 1e and 1f).

The impacts of land-sourced contaminants in coastal regions depend strongly on how those materials are transported in river plumes. Buoyant river plumes have been studied at a wide range of scales, and their response to a range of discharge rates, wind speeds and directions, and other environmental conditions is relatively well known (Chant, 2011; Garvine, 1995; Hetland, 2005; Horner-Devine et al., 2015; O'Donnell, 2010; Yankovsky & Chapman, 1997). Due to the Coriolis force resulting from Earth's rotation, non-equatorial river plumes carry material in the 'downcoast' direction (in the direction of Kelvin wave propagation, moving with the coast on the right in the northern hemisphere) in a buoyant coastal current (Lentz & Largier, 2006). River plumes also may include an anticyclonic 'bulge' near the river mouth, particularly for larger Froude numbers (ratio of river velocity to gravity current speed, with 1 being a critical value between subcritical and supercritical) and small Kelvin numbers (ratio of river mouth width to deformation radius), indicating the dynamical scale of the plume (Chant, 2011; Fong & Geyer, 2001). Winds modify these patterns, with different effects for downwelling or upwelling favorable winds, which drive onshore or offshore surface Ekman transport, respectively, that is, winds blowing to the west or east for a southward-facing coastline in the northern hemisphere. Downwelling winds tend to cause narrowing of the coastal current with a steepened plume front (Lentz & Largier, 2006). Upwelling wind conditions cause enhanced freshwater storage in a region offshore of the river mouth or along the coast in the 'upcoast' direction opposite the buoyant coastal current (Jurisa & Chant, 2013; Rao et al., 2011). Other plume patterns including bidirectional plumes (Hickey et al., 2005) and cross-shore plumes (Yankovsky et al., 2022) occur under certain combinations of river discharge and wind conditions (Xia et al., 2007, 2010). Ambient coastal currents also act to modify plume turning and augment or the decrease coastal current transport (Fong & Geyer, 2002).

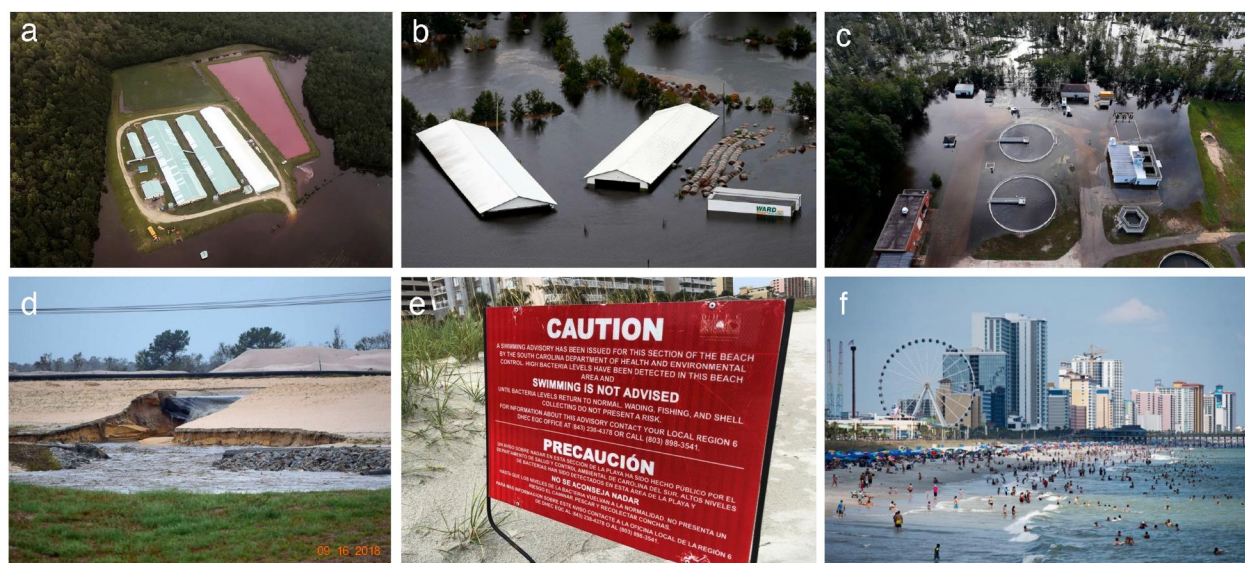


Figure 1. Images of flooding of agricultural and industrial facilities following Hurricane Florence including (a) a hog waste lagoon (Rodrigo Gutierrez/Reuters), (b) poultry operation (Steve Helber/AP), (c) wastewater treatment plant (Gerald Herbert/AP), and (d) coal ash landfill (Waterkeepers Alliance), and beach impacts including (e) sign indicating temporary swimming advisory (WPDE-TV, Florence, SC), and (f) swimmers on a busy weekend at Myrtle Beach, SC (Sean Rayford/Getty Images).

Winds, discharge, and other sources of variability in river plume behavior change rapidly and exceed typical magnitudes during and following extreme storm events. These transient and extreme conditions are not as well understood as steady river plume responses, making it difficult to prepare for river-sourced ocean water quality hazards following storms. Lemagie and Lerczak (2020) studied the response of a river plume to a discharge pulse under downwelling winds, finding that bulge formation was temporarily enhanced at peak discharge, followed by the release and downcoast advection of the freshwater stored in the bulge as discharge decreased. This led to a long tail (extended duration) in the downcoast freshwater transport. Understanding of potential water quality implications of freshwater transport patterns following a discharge pulse can be improved by further study with a range of wind conditions, more extreme discharge rates, and a focus on tracer transport.

Hurricane Florence was a long-track, Cape Verde storm that moved across the tropical Atlantic Ocean over a period of two and a half weeks (Figure 2a). It reached a peak intensity of a Category 4 storm on 11 September 2018, about 3 days before landfall, then weakened substantially before landfall on September 14 (Stewart & Berg, 2019). While winds weakened, the slow speed and hook-shaped path of the storm led to record-breaking rainfall over southeastern North Carolina (Figure 3c), with a maximum four-day accumulation of 91 cm (36 inches) in Elizabethtown within the Cape Fear River Basin. The Cape Fear River Basin is the largest watershed contained entirely within North Carolina, with a drainage area of 23,889 km². The watershed is highly industrialized, with many large cities, wastewater treatment plants, and concentrated animal feeding operations in the floodplain. For example, the Cape Fear River Basin has one of the highest concentrations of concentrated animal feeding operations in the world (C. N. Brown et al., 2020). Several tributaries feed the Cape Fear River, which empties into the South Atlantic Bight near Wilmington, NC. Simulations of the Cape Fear River discharge during Hurricane Florence (Bao et al., 2022; Gori et al., 2020; Yin et al., 2021), consistent with river gauge and water level data, indicate that compound flooding occurred in the estuary during the storm and that peak discharge lagged peak rainfall and winds by several days. Models and satellite imagery have shown that the Cape Fear River plume forms a range of patterns dependent on winds, river discharge, and other factors (Lewis et al., 2019; Xia et al., 2007, 2010), however the response to extreme discharge events for a range of conditions has not been studied in detail.

During Hurricane Florence, severe flooding of agricultural and industrial sites including hog farms and municipal wastewater treatment facilities led to the release of various contaminants into the Cape Fear watershed. Satellite observations indicate that coastal ocean turbidity was strongly enhanced following the event (Lewis et al., 2019). Decisions about local swimming and shellfishing advisories usually rely on intermittent water quality sampling and often focus on local sources including stormwater drainage. However, far-field land-sourced contaminants can also be a substantial source of hazard and have important human and ecosystem health impacts. For example, news reports indicate that in Myrtle Beach, SC, approximately 100 km west (downcoast) of the Cape Fear River, a swimming ban was lifted on September 18, after rainfall subsided, based on local water sampling results (Weissman, 2018). The beach was closed again on September 28 after a subsequent water quality measurement was poor (Kummerer, 2018; Still, 2018; WPDE, 2018). The reason for the later poor water quality measurement is not known, but is consistent with a delayed arrival of contaminants that had transited long distances through the Cape Fear watershed and river plume, reaching the area at an unknown date between September 18 and 28. In addition, water quality sampling often focuses on a small set of indicators, including fecal indicator bacteria, which are sometimes an insufficient measure of potential hazards that persist in the water column for longer timeframes than these more short-lived indicators (Boehm et al., 2009, 2018). Improved understanding of the pathways of land-sourced contaminants in the ocean would aid managers in planning for the onset and duration of coastal water quality hazards.

In this paper, we describe the behavior of tracers representing contaminants in the Cape Fear River plume following Hurricane Florence and for a suite of river discharge and wind scenarios. The ocean modeling approach with river discharge and tracers is described in Section 2. In Section 3, the results of the Hurricane Florence simulation and other scenarios are described and compared to evaluate the sensitivity of patterns of coastal contaminant concentrations to river discharge and winds. The results are discussed in Section 4 and the conclusions are summarized in Section 5.

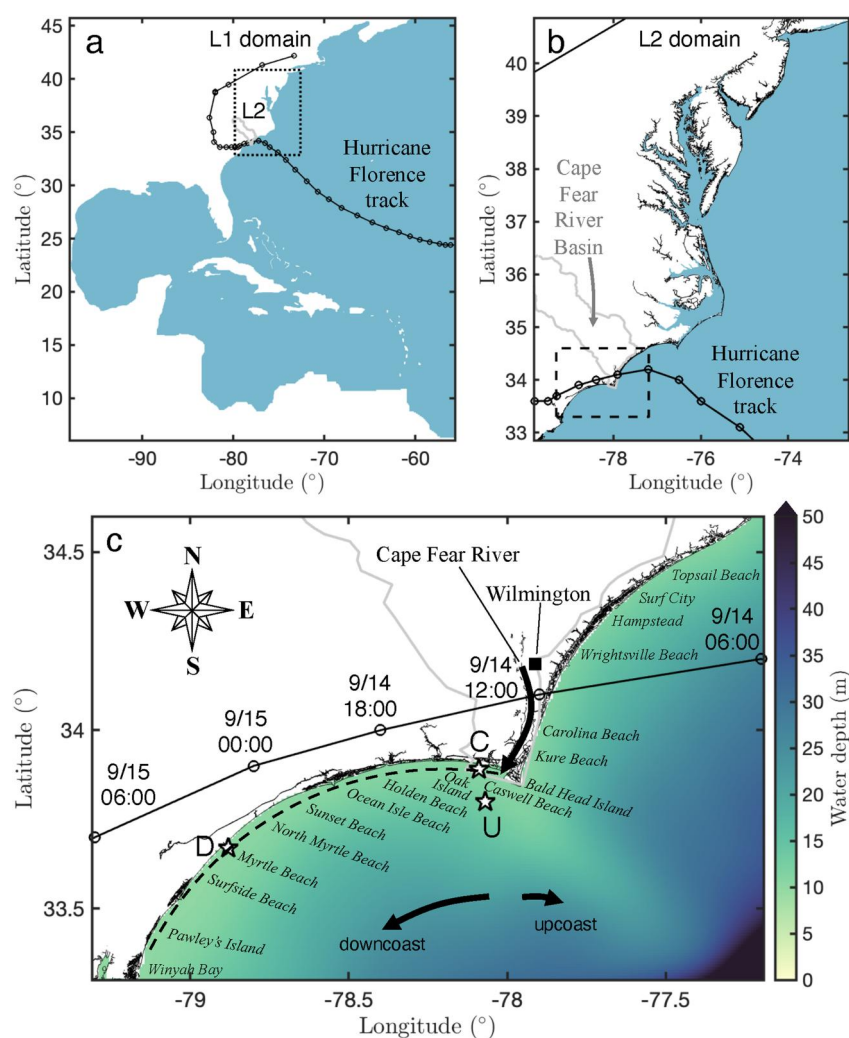


Figure 2. Cape Fear River study region overview. (a) L1 model domain used to generate boundary conditions for L2 domain (dotted box), (b) L2 model domain (dashed box shows zoom to panel c), (c) subset of L2 model domain with the Cape Fear River (center), bathymetry (color scale at right), and locations for time series results (stars: D is a 'downcoast' location far to the west of the river mouth, C represents the 'coastal current' just west of the river mouth, and U is a point in the 'upcoast' or river-plume bulge region to the south or east of the river mouth; arrows indicate downcoast and upcoast directions) versus Latitude and Longitude (compass in panel c). Hurricane Florence track (solid line with circles every 6 hr, times labeled in panel c) and Cape Fear River Basin watershed boundary (gray line) shown in each panel. In (c), the dashed black line is an along-coast transect used to track hazard onset and duration at beaches to the west of river mouth.

2. Methods

2.1. Ocean Model Simulation of Hurricane Florence With Cape Fear River Freshwater and Tracer Source

The evolution of the Cape Fear River plume and passive tracers representing land-sourced contaminants (Section 2.2) for Hurricane Florence and a suite of scenarios (Section 2.3) are simulated using the Regional Ocean Modeling System (ROMS) (Haidvogel et al., 2008; Shchepetkin & McWilliams, 2005), a free surface, terrain-following numerical model that solves the three dimensional primitive equations using the hydrostatic and Boussinesq approximations. ROMS was run within the Coupled Ocean Atmosphere Wave Sediment Transport (COAWST) modeling system (Kumar et al., 2012; Warner et al., 2008, 2010). ROMS has been used extensively to simulate tropical storms and hurricanes on the eastern seaboard and elsewhere (Defne et al., 2022; Hegermiller et al., 2019; Olabarrieta et al., 2012; Warner et al., 2008; Zambon et al., 2014, 2021). The model configurations and parameters used here are based on related previous work in Zambon et al. (2021), in which simulations of Hurricane Florence were evaluated with observations of the storm track and in situ and remotely sensed ocean sea surface temperatures. These results indicate that this model configuration is reasonable to simulate physical processes during this event. However, a major limitation of this

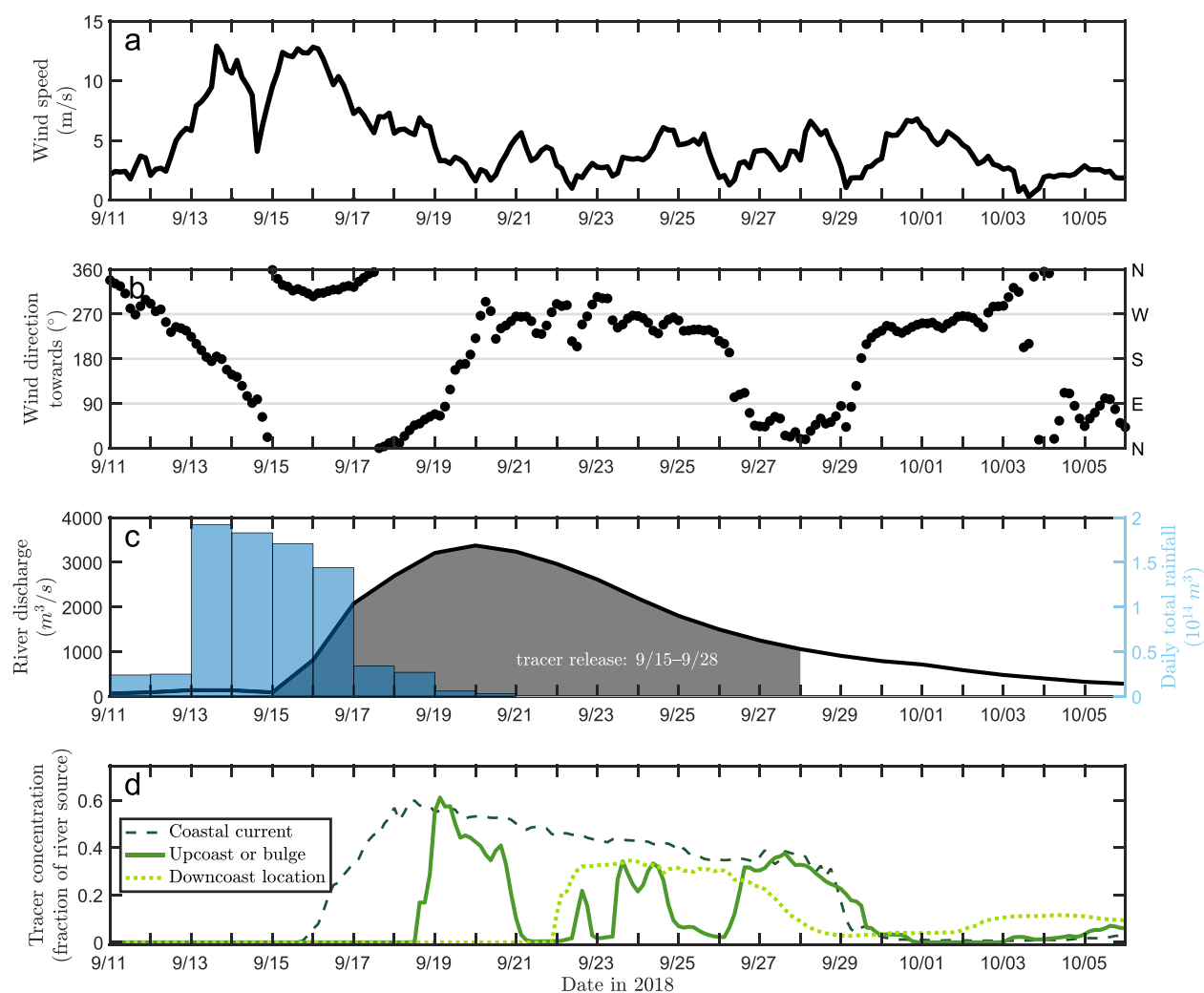


Figure 3. Time series of Hurricane Florence simulation forcing and results. (a) Wind speed near the river mouth (m/s), (b) wind direction ($^{\circ}$ toward; 0° is to N, 90° is to E, 180° is to S, and 270° is to W) (c, left axis) Cape Fear River discharge (m^3/s) with shading indicating time period of tracer release and (c, right axis) daily storm total rainfall volume (10^{14} m^3) from NOAA AHPS (Zambon et al., 2021), and (d) tracer concentrations (fraction relative to the river source) near the river mouth in the coastal current, upstream of the river mouth or in river-plume bulge, and at an example downcoast location approximately 100 km west of the river mouth, near Myrtle Beach, SC (Figure 2c shows locations: C is coastal current, U is upcoast or bulge, and D is the downcoast location).

model study is that we lack observations of the details of the plume salinity or contaminant concentrations to evaluate the model performance in these regards. Thus we chose an appropriate focus for this study to investigate the broad spatial patterns and approximate temporal evolution of river-sourced contaminants under varying discharge and wind conditions. Future observational studies are needed to expand the usability of these model results.

ROMS was first run on a U.S. East Coast domain (L1) with 4-km resolution and 50 vertical layers spanning from 12°N to 46°N and -98°E to -55°E (Figure 2a) for Hurricane Florence conditions, similar to Zambon et al. (2021). This run was used to generate boundary conditions for runs on a smaller model domain (L2), extending from approximately 32.8°N to 40.9°N and -79.8°E to -72.6°E , from northern South Carolina to New York (Figure 2b). The L2 grid has 800-m horizontal resolution ($900 \times 1,000$ grid-points) and 50 vertical layers. For both the L1 and L2 grids, the vertical coordinate is surface intensified in deeper waters and transitions to depth-uniform in depths less than 200 m. Downscaling from a larger L1 domain to versions of the L2 domain has been used widely for applications to the mid-Atlantic region. Lateral east and south boundary conditions in the L2 runs are Chapman explicit for the free surface, Schepetkin for 2D momentum, Gradient for mixing TKE, and Radiation with Nudging for 3D momentum, temperature, and salinity, with closed boundaries on the west and

Table 1

Summary of Regional Ocean Modeling System (ROMS) Ocean Model Runs Including Date Ranges, River Discharge, and Wind Forcing. 'Hurricane Florence' River Discharge Refers to the Detided Discharge Time Series Simulated by Coupled ROMS and WRF-Hydro Simulations for September and October 2018 Conditions, Which had a Peak of 3,377 m³/s on 9/19–9/20 (Figure 3c). 'Hurricane Florence' Wind Forcing Refers to Gridded ERA5 Wind Forcing for September and October 2018 (Figures 3a and 3b)

Run name	Dates (2018)	River discharge	Wind forcing
Florence	09/11–10/06	Hurricane Florence, peak 3,377 m ³ /s	Hurricane Florence
River-base	09/15–10/06	Baseflow condition, 100 m ³ /s	Hurricane Florence
River-25%	09/15–10/06	25% Hurricane Florence, peak 884 m ³ /s	Hurricane Florence
River-50%	09/15–10/06	50% Hurricane Florence, peak 1,688 m ³ /s	Hurricane Florence
River-200%	09/15–10/06	200% Hurricane Florence, peak 6,754 m ³ /s	Hurricane Florence
Wind-N	09/17–10/06	Hurricane Florence	constant, 5 m/s to N
Wind-E	09/17–10/06	Hurricane Florence	constant, 5 m/s to E
Wind-S	09/17–10/06	Hurricane Florence	constant, 5 m/s to S
Wind-W	09/17–10/06	Hurricane Florence	constant, 5 m/s to W

Note. Time series of synthetic forcing conditions are shown in Figures 6a and 8a.

north where the boundary grid points are all land cells (Kumar et al., 2012). Harmonic viscosity was set to 0.5 m²/s and the GLS turbulence closure scheme was used. Atmospheric forcing was derived from a 30-km ERA5 product (Hersbach et al., 2020; Li et al., 2022) (Figures 3a and 3b), which is similar to atmospheric fields from higher resolution coupled WRF-ROMS simulations of Hurricane Florence (Zambon et al., 2021), and expected to be reasonable to simulate the bulk response of the river plume to wind forcing. The model was run with a baroclinic timestep of 30 s simulating the time period from September 11 to 6 October 2018 (Table 1). A complete description of model parameters can be found in the data archive (see Data Availability Statement).

The Cape Fear River discharge was implemented as a point source at one horizontal grid cell at the coast with uniform vertical distribution (Figure 2c) using a detided (33-hr filter) discharge timeseries extracted from a coupled ocean (ROMS) and hydrologic (WRF-Hydro) (Gochis et al., 2018) model simulation of the Hurricane Florence event (Bao et al., 2022) (Figure 3c). In contrast to using stream-gauge observations from a more inland location, using river discharge from a point near the river mouth in a hydrologic model has several potential advantages. First, the hydrologic model includes inputs from additional tributaries and overland sources near the river mouth that are not included in an upstream gauge measurement. In addition, in the work of Bao et al. (2022), WRF-Hydro was two-way coupled with a depth-averaged ROMS model configuration and was thus able to simulate the effect of interacting ocean and river processes on discharge and compound flooding. Finally, while forcing with stream-gauge data can be suitable for hindcasts, the approach here is a step towards future studies extending this work to forecast scenarios that require the simulation of rainfall-runoff process with a hydrological model. Given the model resolution and coastal river point source location, estuarine adjustment is not resolved here, which may lead to errors in the salinity and tracer concentrations and vertical structure in the river outflow. Thus, here we focus on the response of coastal plumes for given fixed water properties at the Cape Fear River mouth based on climatology (salinity: 0.5 psu, temperature: 24.5°C, tracer concentration: 1) and a constant vertically uniform distribution consistent with a strongly forced (river-dominated) system during a large discharge event. The results are not sensitive to small changes in the temperature, salinity, and vertical structure of the river source. Other rivers in the model domain were implemented as point sources from the National Water Model (Blodgett, 2019), and contributed to extreme freshwater discharge in locations such as Pamlico Sound, however interaction of these rivers with the Cape Fear River plume is not expected to significantly affect the results.

For this study of the post-storm plume behavior and the related potential for water quality hazards along the coast, we did not couple with a surface wave model. Large surface waves (wave height ~5 m) were present and drove elevated coastal water levels and ocean mixing during the storm (9/13–9/17) (Zambon et al., 2021). This was separated in time from the elevated river discharge following the storm (9/17–9/27) (Figure 3), when surface waves were observed to be small (wave height ~1 m) and thus not expected to have a substantial impact on the large-scale behavior of the river plumes. Near the beach, detailed pathways of contaminants could be modeled

more accurately with a higher resolution model with surface wave coupling (Brasseale et al., 2023; Feddersen et al., 2021) to simulate rip currents and other nearshore processes that affect the longshore transport, dilution, and cross-shore exchange of material (Moulton et al., 2023).

2.2. Tracking Land-Sourced Contaminants

To study the evolution of hypothetical land-sourced contaminants, a passive non-decaying conservative tracer with a concentration of 1 is released at the Cape Fear River point source. Horizontal and vertical tracer advection are solved with the HSIMT-TVD algorithm (Wu & Zhu, 2010). A finite duration tracer release is performed to approximate the release of contaminants during a flooding event. The release begins at 00:00 UTC on 9/15/2018, approximately when elevated discharge began entering the Cape Fear area, following inundation and contaminant release at upstream sites. The release ends at 00:00 on 9/28/2018, based on declining water levels and discharge in the Wilmington area and news reports of when floodwaters receded. Results are similar to a continuous dye release (not shown), in which coastal ocean dye concentrations are low prior to 9/15 and after 9/28 when discharge rates were low, with higher dye concentrations near the end of the simulation after the date of the end of the finite dye release.

Surface concentrations of dye are the primary focus of this study, as they tend to be highest in the buoyant plume, representing the maximum hazard. Spatial patterns of the maximum potential contaminant concentrations have clear implications for planning related to human and ecosystem exposures, whereas other metrics such as net transport or depth-averaged concentrations typically have less relevance to management decisions. In addition to studying complex spatial patterns of the tracer, timeseries of tracer concentration are investigated at three locations representing the initial formation of the coastal current just downcoast (west) of the river mouth, a river-plume bulge or upcoast region (offshore or east), and a far downcoast (west) location, Myrtle Beach, SC (symbols in Figure 2c). Results are similar to selecting nearby points or averaging over broader regions within several km. The 'upcoast or bulge' region encompasses contaminants that are transported south or east of the river mouth via a classic dynamical river-plume bulge, along with wind-driven transport, wind modification of bulge circulation, and other processes. Myrtle Beach, SC was selected as an example of a popular beach that may be affected by a relatively distant river under high discharge conditions; results for other locations west of the river mouth are qualitatively similar and are discussed via maps and an along-coast transect.

To assess the onset and duration of water quality hazards, a hazard threshold level of 5% of the riverine input value (a concentration of 0.05) is chosen to approximately represent a pathogen or contaminant level that may cause human illness, or nutrient or fine sediment levels that could cause harmful ecosystem impacts. Prior studies have selected a threshold based on a percentage concentration relative to undiluted effluent that has a significant likelihood of causing human illness, for example, for a specific pathogen such as norovirus (Brasseale et al., 2023; Feddersen et al., 2021). Here, the riverine concentration relative to undiluted sewage or contaminants released upstream is not known, and thus the threshold chosen is somewhat arbitrary. Results for onset of hazard are not highly sensitive to varying this choice over a wide range of values (0.01%–20%), due to steep fronts associated with initial contaminant arrival to a given location. However, the duration of hazard is more sensitive to this choice, given the more slowly decaying concentrations as contaminants dilute. This study is intended to demonstrate how hazard onset and duration can be quantified for a given threshold, and the approach could be applied in future studies that target a specific contaminant with a constrained source concentration and exposure threshold.

A conservative passive tracer study was chosen as a starting point for tracking contaminant pathways, thus this work is most applicable to tracers with behavior timescales that are long or similar to the simulated days to weeks timescales of hazard onset and duration. Future work could consider the behaviors of coastal contaminants, including biogeochemical cycling, settling rates, and pathogen mortality due to sunlight exposure or other factors (e.g., some common pathogens have e-folding time scales of days to months) (Boehm, 2003; Boehm et al., 2018).

2.3. Scenarios Motivated by Changing Weather, Climate, and Land Use

In addition to simulating the Hurricane Florence event, a suite of eight semi-idealized scenarios were simulated to investigate the sensitivity of the Cape Fear River plume and contaminant transport to changing conditions (Table 1). To represent a range of precipitation conditions including future extreme events, the Hurricane Florence simulation setup (Section 2.1) was modified to adjust the discharge rate beginning at 00:00 on 9/15 to

either a constant baseflow condition of $100 \text{ m}^3/\text{s}$ or a percentage or multiple of the Hurricane Florence discharge curve (Table 1, Figure 6a). Over the past several decades, events in the Cape Fear River Basin with approximately 50% of the Hurricane Florence discharge have occurred every several years, and events with around 25% of the Hurricane Florence discharge have happened on closer to an annual basis (Yin et al., 2021). High discharge events in the region have been increasing (Paerl et al., 2019), and thus the 25% and 50% events may represent events that occur more frequently in the future. In addition, a scenario with 200% the Florence discharge is selected as an extreme future scenario. Based on prior simulations of two tributaries in the Cape Fear River Basin, an event with 200% of the Hurricane Florence peak discharge could be expected to occur for a rainfall event with approximately 40% more precipitation than the Hurricane Florence event (Yin et al., 2022). This percentage increase is in line with the average increase in extreme precipitation projected for the RCP 8.5 emissions scenario, or 4°C warming by 2100 (Intergovernmental Panel on Climate Change, 2021).

The post-storm wind speeds following Hurricane Florence were moderate at around 2–6 m/s near the river mouth, and the wind direction varied over several-day timescales (Figures 3a and 3b). To understand the response of the plume to the varying wind direction observed following Hurricane Florence, as well as for different future post-storm weather patterns, four cases were run with winds set to a constant speed (5 m/s) and one of four constant directions beginning at 00:00 on 9/17 (Table 1, Figure 8a). For the synthetic wind cases (Section 2.3), boundary conditions are derived from the U.S. East Coast domain (L1) Hurricane Florence simulation and thus these simulations include the influence of the larger scale currents in addition to the modified local wind forcing. Based on a case with zero winds (not shown), this effect is expected to be small. Discharge rate and wind direction were selected here as examples of strong controls on river plume behavior. Future work could investigate additional synthetic scenarios that represent other aspects of changing climate, weather, and land use, including a range of timing of peak discharge, timing and magnitude of contaminant inputs, wind speeds and patterns, tidal processes, the behavior of the Gulf Stream and other larger scale ocean currents, and the variation of the mixed layer depth and other ocean state characteristics.

3. Results

3.1. Hurricane Florence

The elevated discharge associated with Hurricane Florence ('Florence' case in Table 1; Figure 3c) began ramping up from 9/15–9/17, several days after the storm made landfall and precipitation peaked. Initially, a westward buoyant coastal current formed (Figures 4a, 4e, 4i and 4m, dark green dashed curve in Figure 3d) with contaminant (tracer) concentrations around 50% of the value at the river mouth, salinity near 15–25 psu, slightly cooler temperature than the ambient sea surface temperature, and westward ocean currents due to a downwelling wind. The coastal current was relatively thick and narrow with a steepened plume front, consistent with the moderate to strong downwelling wind (Lentz & Largier, 2006).

Between 9/17–9/19, the discharge peaked and the downwelling winds transitioned to upwelling winds (Figure 3). During this time, the plume became bidirectional (Figures 4b, 4f, 4j and 4n, dark green dashed curve and green solid curve in Figure 3d). The westward coastal current widened, shallowed, and partially separated from the coastline, consistent with prior studies of plumes for upwelling or offshore winds (Fong & Geyer, 2001; Jurisa & Chant, 2013; Lentz, 2004). The eastward extension of the plume resulted from a combination of advection by eastward currents driven by the upwelling favorable wind (Figure 4n) and the accumulation of freshwater in an anticyclonic river-plume bulge under the higher discharge levels (Lemagie & Lerczak, 2020).

After the peak discharge, the bulge concentration decreased (green solid curve in Figure 3d), and previously stored freshwater was advected slowly downcoast, augmented by forcing by downwelling favorable winds (Figures 4c, 4g, 4k and 4o). Around this time, more than a week after the storm arrival and 4 to 5 days after the storm had passed, a coastal current with elevated contaminant concentrations reached locations far downcoast like Myrtle Beach, SC (increase in light green dotted curve on 9/22 in Figure 3d). Several cycles of alternating wind directions led to intermittent storage in and release from the upcoast or offshore bulge region (Figures 3b and 3d).

Finally, discharge decreased toward baseflow conditions and the finite duration tracer release ended (Figure 3c). The last pulse of contaminants released from the river was advected downcoast in the coastal current with decreasing concentration as the discharge decreased and the plume was diluted (Figures 3d, 4d, 4h and 4p). In addition, a final lobe of contaminated freshwater stored near the mouth was advected downstream by the ambient

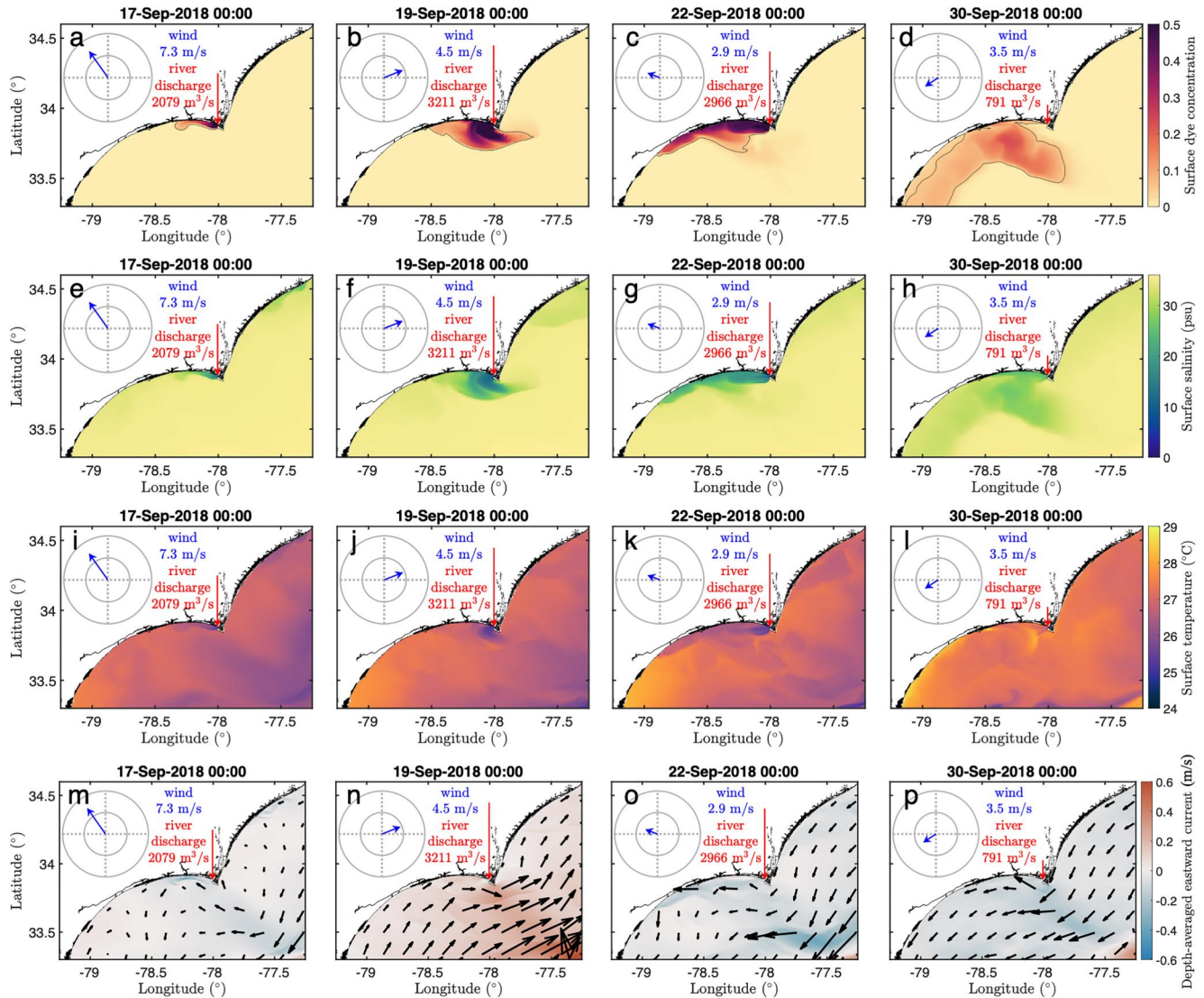


Figure 4. Plan view surface dye concentration relative to the riverine concentration ((e), (f), (g), (h) first row, contour shows plume edge defined as 5% of the riverine concentration), surface salinity ((i), (j), (k), (l) second row), surface temperature ((m), (n), (o), (p) third row), and depth-averaged currents ((m), (n), (o), (p) fourth row, arrows plotted every 20 grid cells, color scale shows east-west component) for four times (00:00 on 17-Sep: first column, 19-Sep: second column, 22-Sep: third column, and 30-Sep: fourth column) near the Cape Fear River mouth during the Hurricane Florence simulation versus Latitude and Longitude. Wind speed and direction (blue vector with gray circles at 5 and 10 m/s, and blue text), and river discharge (red arrow and text) are indicated in each panel.

ocean currents (peak in green solid curve near 9/27–9/29 in Figures 3d, 4d, 4h and 4p). Contaminant concentrations remained elevated at the downcoast Myrtle Beach location beyond the end of the simulation on 10/6, nearly 3 weeks after Hurricane Florence made landfall and more than 2 weeks after the storm had passed (light green dotted curve in Figure 3d).

3.2. Response to Varying River Discharge

Hypothetical storm events similar to Hurricane Florence with different peak discharge rates have similar spatial patterns of tracer concentrations (Figure 6a; ‘River’ cases in Table 1). Similar to the Hurricane Florence case, cases with peak discharge of 200%, 50%, and 25% the Hurricane Florence discharge rate initially formed a westward coastal current (Figures 5e, 5i, 5m). As downwelling winds transitioned to upwelling winds and discharge simultaneously increased, the plume for these cases included a bulge or other eastward extension (Figures 4b, 5f, 5j and 5m). When upwelling winds relaxed or transitioned back to downwelling winds, material

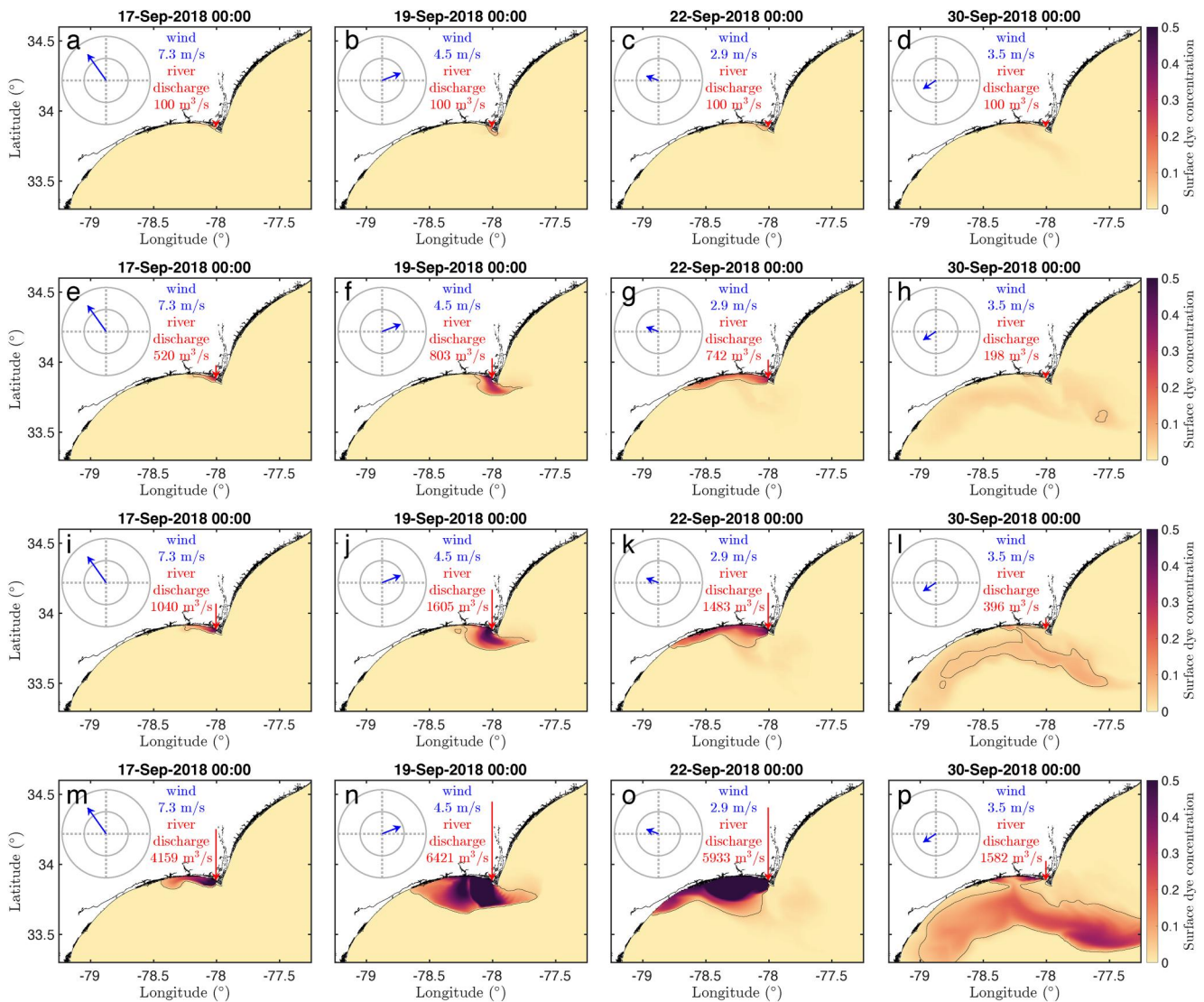


Figure 5. Plan view surface dye concentration relative to the riverine concentration for four discharge cases (a), (b), (c), (d) top row, constant baseflow condition; (e), (f), (g), (h) second, (i), (j), (k), (l) third, and (m), (n), (o), (p) fourth rows: 25%, 50%, and 200% the Hurricane Florence peak discharge respectively) at four times (00:00 on 17-Sep: first column, 19-Sep: second column, 22-Sep: third column, and 30-Sep: fourth column) near the Cape Fear River mouth versus Latitude and Longitude. Wind speed and direction (blue vector with gray circles at 5 and 10 m/s, and blue text), and river discharge (red arrow and text, note: scale in last row is half of the other panels) are indicated in each panel. Contour shows plume edge defined as 5% of the riverine concentration.

that was stored upcoast was released (Figures 4b, 5g, 5k and 5o). As discharge returned to baseflow conditions and the tracer release ended, the contaminants were advected downcoast and diluted (Figures 4c, 5h, 5l and 5p). The width of the coastal current and the distance of eastward and offshore extension were larger for larger discharge (Figure 4 top row and 5). The only case with a qualitatively different pattern was the case with the weak baseflow discharge level ('River-base' in Table 1), for which the plume salinity anomaly was so low that a coastal current was weak and mostly confined within 10 km of the river mouth (Figures 5a–5d).

The storm discharge cases with 200%, 50%, and 25% of the Hurricane Florence peak discharge differ in several key ways, including the arrival times and concentration magnitudes of the tracer representing land-sourced contaminants. The arrival of the coastal current immediately downcoast of the river mouth and the development of a bulge or upcoast storage region occurs most quickly and with highest concentration for larger discharge values (Figures 6b and 6c). Similarly, for the larger discharge values, tracers have higher concentration and arrive sooner to the downcoast Myrtle Beach location, consistent with a faster nose propagation speed for a coastal current with a stronger density anomaly (Figure 6d). For the case with 200% of the Hurricane Florence discharge,

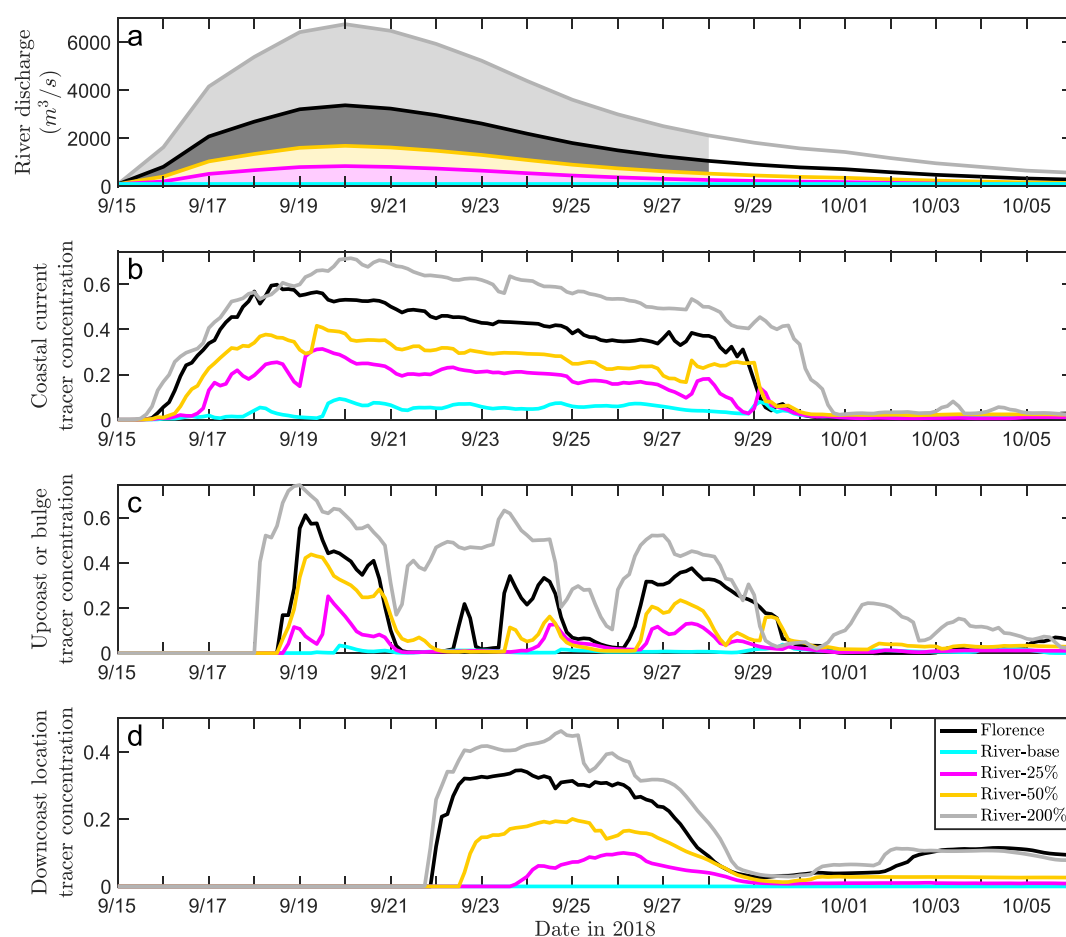


Figure 6. Time series of results for synthetic discharge cases (Table 1) for a constant baseflow condition or proportional to the Hurricane Florence discharge record. (a) Cape Fear River discharge (m^3/s) for Hurricane Florence (black), a baseflow condition of $100 \text{ m}^3/\text{s}$ (cyan), and 25% (magenta), 50% (gold), and twice (gray) the Hurricane Florence discharge (key in panel d), with shading indicating time period of tracer release, and tracer concentrations (fraction of river source) at three locations: (b) in the coastal current just west of the river mouth, (c) upcoast of the river mouth or in an offshore bulge, and (d) at an example downcoast location near Myrtle Beach, SC (Figure 2c shows locations: C is coastal current, U is upcoast or bulge, and D is the downcoast location).

tracers arrive in the morning on 9/21, several hours earlier than for the Hurricane Florence case. For 50% and 25% of the Hurricane Florence discharge, tracers arrive midday on 9/22 and 9/23, respectively, more than a week after the storm made landfall and nearly a week after the large precipitation rates subsided. For the case with baseflow discharge levels, tracers never exceed the threshold at the downcoast location (Figure 6, cyan curves). Downcoast tracer concentrations for all of the non-baseflow cases begin to decrease around 9/27–9/29, shortly after the tracer release ended as discharge subsided.

3.3. Importance of Post-Storm Wind Direction

Synthetic scenarios with different constant and uniform post-storm wind directions suggest that the pathways of freshwater and tracers are highly sensitive to post-storm wind patterns ('Wind' cases in Table 1, Figure 7). For all wind directions (Figure 8a), the tracer concentration in the coastal current just west of the river mouth is similar, representing the river source prior to modulation by winds and Earth's rotation (Figure 8b). The downcoast and upcoast behaviors vary widely (Figures 7, 8c and 8d). For downwelling favorable winds to the west, there is a more rapid arrival of tracer at Myrtle Beach, consistent with a more confined and thicker coastal current (Figure 7m,n,o, red solid curve in Figure 8d) (Fong & Geyer, 2001; Lentz & Largier, 2006). Similarly, for offshore-directed winds to the south, tracers arrive at Myrtle Beach around 9/19, just a few days following the

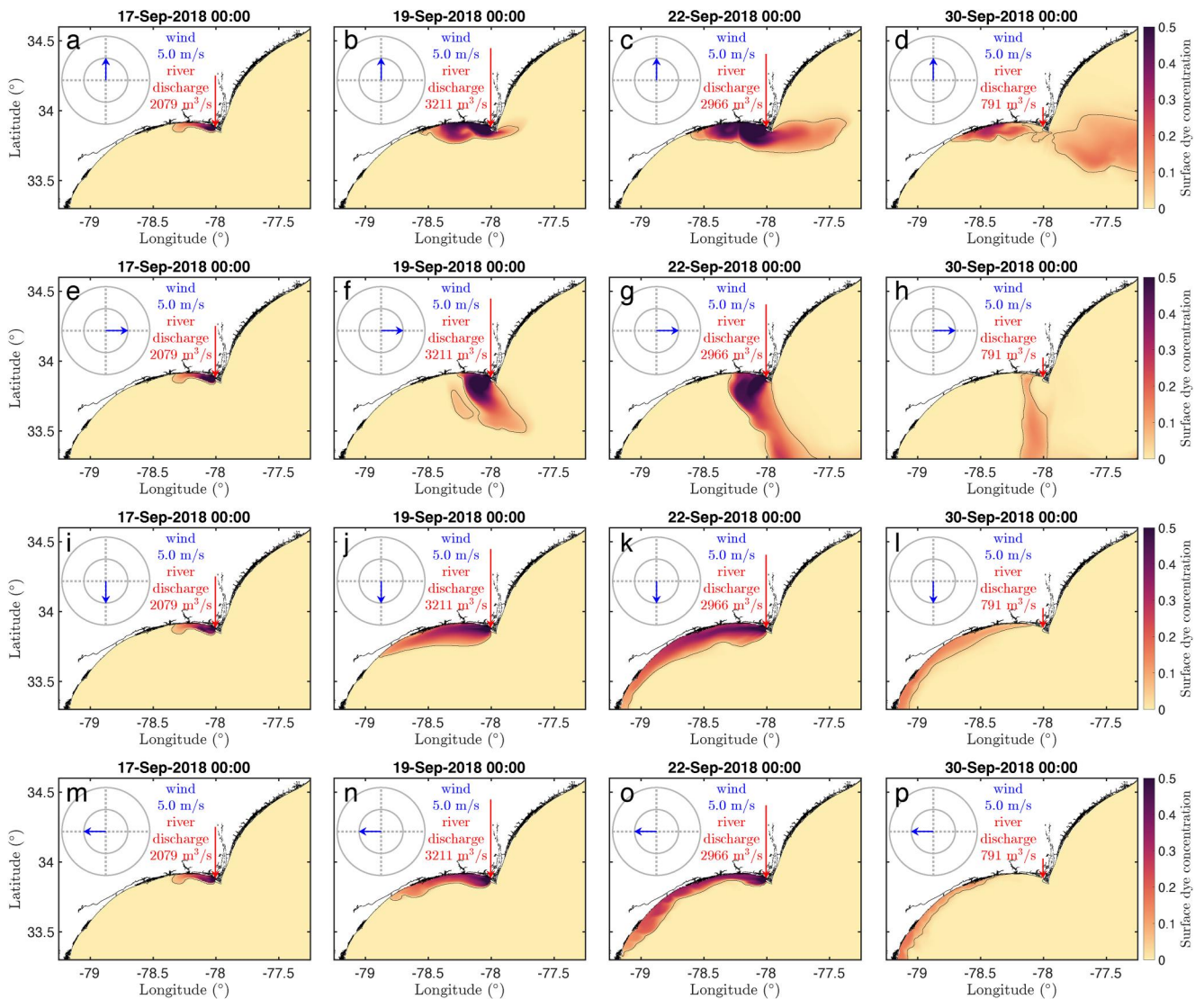


Figure 7. Plan view surface dye concentration relative to the riverine concentration for cases with constant 5 m/s wind speed and four directions ((a), (b), (c), (d) top row, to N; (e), (f), (g), (h) second row, to E; (i), (j), (k), (l) third row, to S; (m), (n), (o), (p) fourth row, to W; all 5 m/s) at four times (00:00 on 17-Sep: first column, 19-Sep: second column, 22-Sep: third column, and 30-Sep: fourth column) near the Cape Fear River mouth versus Latitude and Longitude. Wind speed and direction (blue vector with gray circles at 5 and 10 m/s, and blue text), river discharge (red arrow and text), and locations of coastal current, upcoast/bulge, and downstream locations (stars corresponding to locations in Figure 2c) are indicated in each panel. Contour shows plume edge defined as 5% of the riverine concentration.

storm (Figures 7i–7k, purple dotted curve in Figure 8d). For the southward wind case, there are relatively low concentrations of tracer in a small bulge near the river mouth that only forms near peak discharge on 9/20 (Figures 7j, 7n and 8, and red and purple curves in Figure 7).

In contrast, for upwelling favorable winds to the east, there is no coastal current that reaches Myrtle Beach and there are substantial tracer concentrations in a large bulge feature (Figures 7f and 7g, solid blue curves in Figures 8c and 8d). For onshore winds to the north, the plume has a bidirectional pattern, with tracer traveling 10s of km west nearly reaching Myrtle Beach, and also substantial quantities of tracer stored near or east of the river mouth (Figures 7b and 7c, dashed light blue curves in Figures 8c and 8d). The behavior of the plume for the Hurricane Florence case shows most similarities to the westward or southward winds, where the tracer is primarily transported in a westward coastal current (Figure 4, top row). Finally, for the westward and southward wind cases and the Hurricane Florence winds, contaminants at Myrtle Beach continue to exceed 5% of the riverine

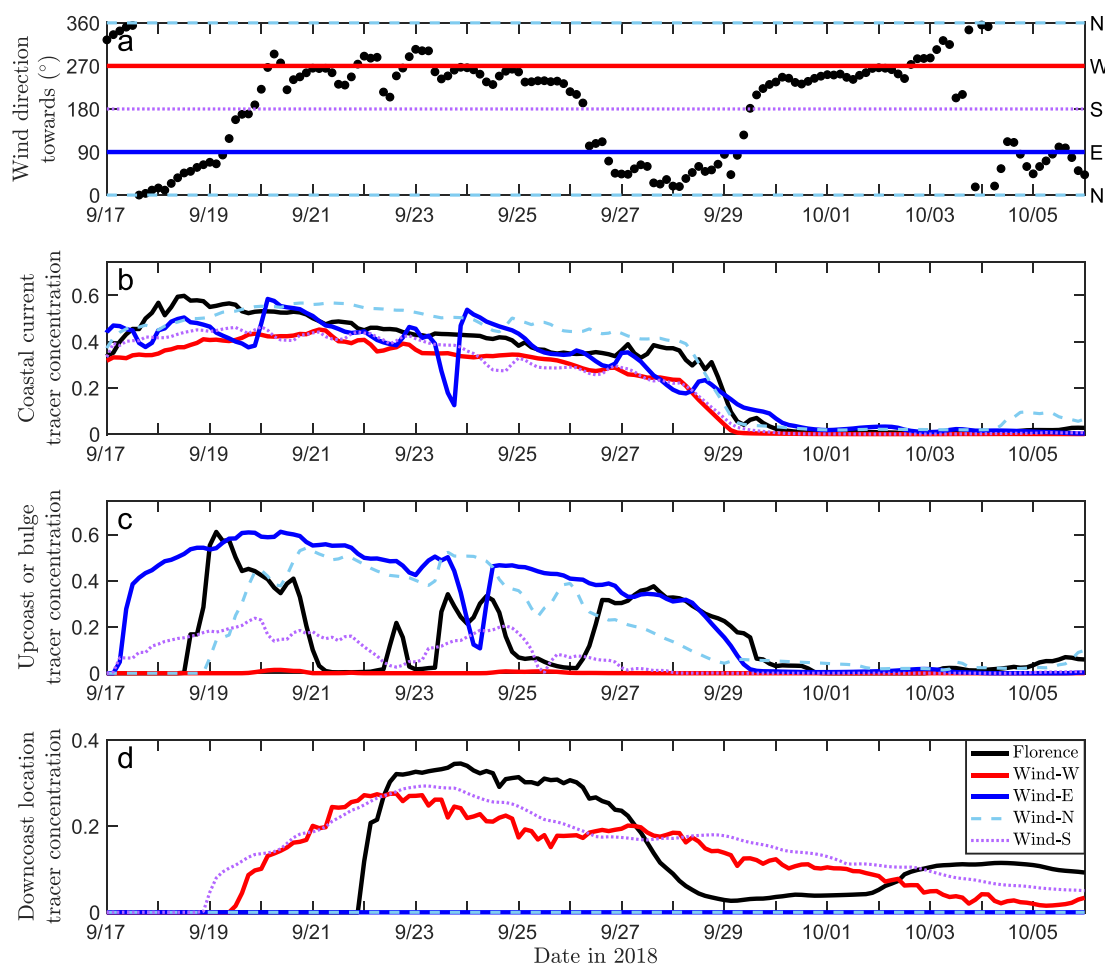


Figure 8. Time series of results for synthetic wind cases (Table 1) with constant post-storm wind direction and speed 5 m/s. (a) The wind direction for synthetic cases with winds to the east (blue solid), to the south (purple dashed), to the west (red solid), and to the north (light blue dashed), and tracer concentrations (fraction of river source) at three locations: (b) in the coastal current just west of the river mouth, (c) upcoast of the river mouth or in an offshore bulge, and (d) at an example downcoast location near Myrtle Beach, SC (Figure 2c shows locations: C is coastal current, U is upcoast or bulge, and D is the downcoast location).

concentration weeks after the storm has passed and days after the contaminant release ended, suggesting that material stored near the mouth was eventually advected downcoast (Figures 4d, 7l, 7p and 8d).

4. Discussion

4.1. River Plume Dynamics Contribute to a Delay in Onset and an Extended Duration of Water Quality Hazard

The exposure of a particular coastal location to contaminated freshwater depends on a number of factors: the magnitude, spatial distribution, and timing of precipitation; upstream sources of contamination; watershed characteristics; resulting discharge at the river mouth; river plume dynamics; ambient ocean conditions; and atmospheric forcing on the ocean. Our study investigates contaminant exposure of the coastal region within ~100 km of the river mouth for varying peak discharge rates and post-storm wind patterns. Results suggest that river plume dynamics including coastal current propagation, bulge formation, and modulation by winds can lead to both a temporal delay and an extended duration in hazard exposure at beaches downcoast (here, west) of the river mouth.

Due to the timescales of hydrologic processes, the speeds of river-plume coastal currents, and river plume patterns under different discharge and wind conditions (Figure 9), substantial time lags may occur between the time with peak precipitation, the time with peak discharge, and the onset time of hazardous freshwater arrival at a coastal site. For all cases studied here, a westward coastal current formed near the river mouth (Figures 5, 7 and 9). For baseflow

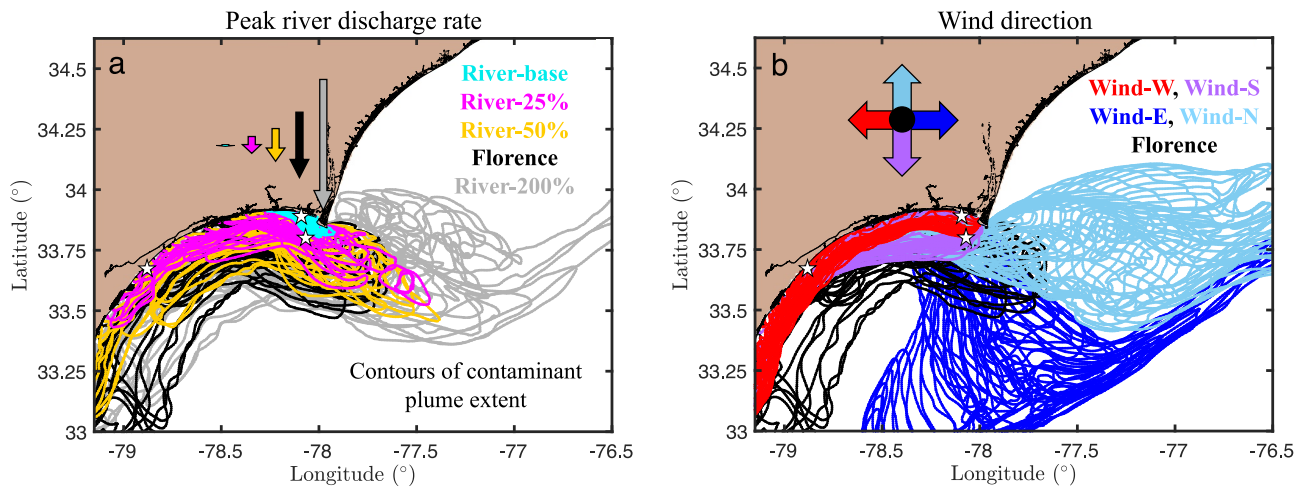


Figure 9. Contours of contaminant plume extent (5% of riverine tracer concentration) every 12 hr from 11-Sep-2018 00:00 to 06-Oct-2018 00:00 for Hurricane Florence simulation (black curves in a and b) and (a) varying discharge rate from a constant baseflow condition of $100 \text{ m}^3/\text{s}$ (cyan) to storm discharge with peak discharge of 25% (magenta), 50% (gold), and 200% (gray) of the Hurricane Florence peak discharge ($3,377 \text{ m}^3/\text{s}$, Table 1), and (b) varying constant wind direction to the west (red), to the south (purple), to the east (blue), and to the north (light blue) with speed 5 m/s after the storm (Table 1).

conditions, the plume Froude number is low enough (less than 0.1) that most freshwater is carried downcoast rather than entering a river plume bulge, however, the small influx of freshwater and tracers are mixed near the river mouth to such a low buoyancy anomaly that no material reaches the downcoast location (Figure 6d). For the other discharge cases, the plume Froude number at peak discharge reaches large enough values (order 1) for this relatively small Kelvin number system (order 0.1–1) that both a coastal current and a river plume bulge form (Avicola & Huq, 2002; Fong & Geyer, 2001; Yankovsky & Chapman, 1997). For the range of discharge rates tested (25%–200%), the first arrivals of contaminants at the downcoast location are between 9/21 and 9/23, around a week after the initial increase in river discharge, consistent with the expected range of nose propagation speeds (order 0.1–1 m/s) for the different buoyancy anomalies associated with percentages of the Hurricane Florence discharge rate (Figure 6d). The presence of downwelling (westward) or seaward (southward) winds leads to a more rapid arrival of contaminants downcoast (Figure 8d), associated with a coastal current that is thicker and confined to a narrower region adjacent to the coast (Figure 9b). The presence of upwelling (eastward) winds, which drive the strongest offshore transport, or onshore (northward) winds, which drive the strongest eastward transport, could cause material stored in this upcoast region to later be transported downcoast (Figure 9b). Notably, none of the conditions studied here led to material reaching the coast north of Cape Fear (Figure 9).

Relative to the temporal pattern of precipitation, which mostly occurred in a 3-day window from 9/14–9/17 (Figure 3c), the exposure of coastal sites to hazardous conditions was extended due to properties of the watershed and river plume and ocean dynamics. Although the simulations performed here were not long enough in duration to see the contaminant levels return to zero, it is clear that under some conditions the duration of contaminant exposure of the downcoast site can be several weeks after the storm arrival, substantially longer than the period of precipitation, and longer than the time period of the finite duration tracer release (see the 9/29 and later time period in Figures 3d, 6d and 8d). This long duration or ‘tail’ in exposure is even more pronounced for cases in which freshwater resided in a bulge or upcoast storage region and was subsequently released downcoast due to decreasing plume Froude number associated with decreasing discharge rate (Figures 3, 6 and 8), or due to a change in wind direction (Figure 3).

For the Hurricane Florence case, the formation of a bulge near the river mouth due to high discharge and upwelling winds initially delayed the downcoast transport of land-sourced material, which was then periodically released due to transitions to downwelling winds and reduced discharge as the flood event subsided (Figures 3 and 4). The periodically released bulge material had a widening footprint and moved downcoast with ambient offshore coastal currents at a slower rate than the buoyant coastal current. After no new contaminants were released, a significant lobe of residual contaminated freshwater was apparent near and offshore of the river mouth, which was released and advected downstream when downwelling winds resumed (Figures 3 and 4).

4.2. Applications to Coastal Management

Decisions about swimming and shellfishing advisories are often made based on water quality sampling, which is typically performed only once to several times per month due to limited resources for sampling and the high cost of laboratory assessments, or based on a fixed duration of several days following rain events. Coupled or linked atmospheric, hydrologic, and oceanic models are capable of simulating the release, downstream transport, and evolution in the coastal ocean of tracers representing contaminants. These complex models can be used as predictive tools for when conditions may be hazardous as well as pinpointing when to take water quality samples (Feddersen et al., 2021; Stone et al., 2022). Additionally, they can be used to guide development of simplified models that are even more readily deployed for planning (Brasseale et al., 2023). These model tools thus can directly help mitigate the information gap resulting from infrequent water quality sampling.

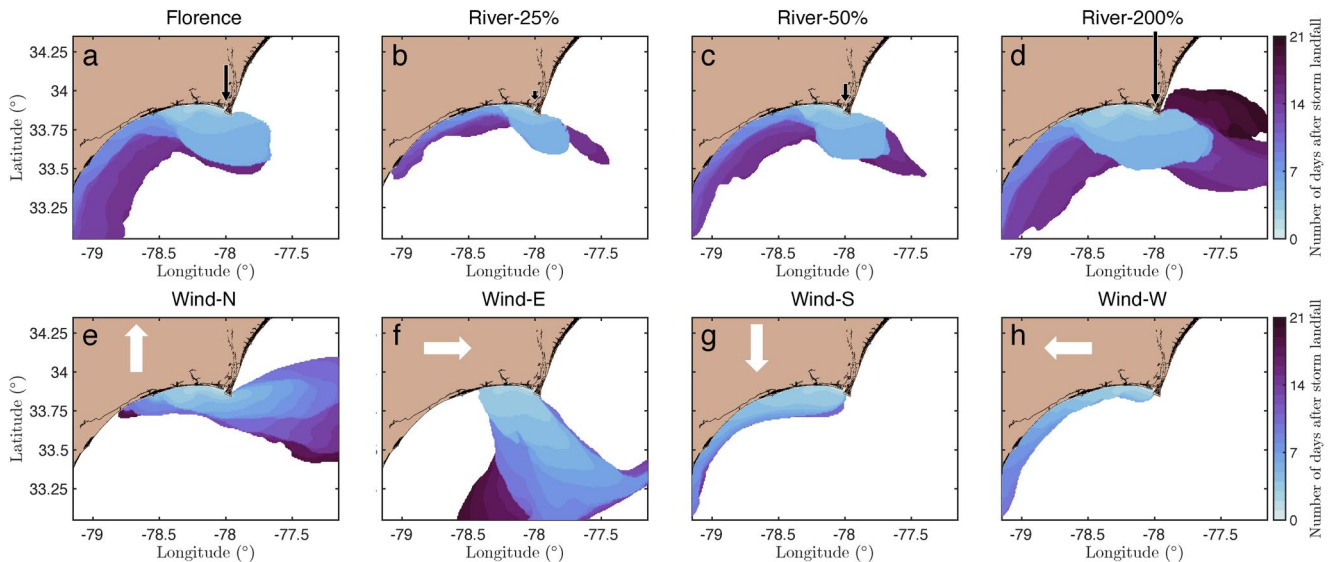
The scenarios considered here based on Hurricane Florence and potential future precipitation events and weather patterns can be presented in a hazard management context using maps of hazard onset and duration, the earliest and latest times when dye exceeds a prescribed threshold (Figure 10). In addition, for assessing hazard onset and duration near shore, these quantities can be extracted from the maps along a transect following the coast (Figure 11, dashed line in Figure 2c). Hazard onset and duration times are often a focus in the natural hazards community as they boil down complex hazard spatiotemporal patterns (Figures 4, 5 and 7) to the key information needed for management: when is a hazard likely to begin and when is it likely to end? These maps and along-coast profiles could be used as a tool for managers in coastal regions to look up how and when their region may be impacted by river-sourced contaminant hazards. For example, Figures 10a, 10i and 11a indicate that under the conditions following Hurricane Florence, beaches to the east of $\sim 78.7^\circ$ West (near the NC-SC border), such as Ocean Isle Beach, NC may have had an earlier onset of water quality hazard, approximately 3–4 days before sites just 10–20 km away to the west, such as North Myrtle Beach, SC. The Longitude location separating the earlier or later hazard onset shifts further west for larger discharge values (Figures 10b–10d and 11b). The presence of this strong spatial gradient in hazard onset occurs for cases with northward, eastward, or variable winds (Figures 10a, 10e, 10f and 11c), whereas westward or southward wind conditions lead to a more smoothly varying and earlier onset of hazard (Figures 10g, 10h and 11c). These trends can supply managers with an understanding of specific scenarios, as well as a range of possible outcomes, and can be a tool to guide preferred times for water quality sampling, for example, avoiding sampling too shortly following a storm when contaminated waters may still be en route.

The results here suggest that when deciding when to lift storm-related swimming advisories, in addition to considering local stormwater sources, municipalities would benefit from considering the arrival of contaminants from distant sources, which may be delayed due to slower pathways of floodwaters through watersheds and river plumes, particularly for variable upwelling wind conditions. As the impacts on beaches at various locations along the coast is highly sensitive to the winds acting on river plumes, it is worth noting that while tropical storm wind forecasts are obviously important, accurate forecasts of wind conditions in the wake of the storm are also an important forecasting priority with respect to water quality hazards. The maps and profiles of hazard onset and duration for given weather conditions (Figures 10 and 11) may be useful to optimize the timing and location of limited resources for nearshore water quality sampling. To extend these tools to other applications including identifying priority sites for removal of waste sources in flood-prone areas, additional climate and land use scenarios could be considered, and the ocean passive tracer studies could be augmented with full coupling with hydrology models (Bao et al., 2022; Ye et al., 2020; Yin et al., 2021) and simulation of biogeochemical constituents. Future work evaluating the hazard onset and duration maps with observations is needed prior to use as a management tool.

5. Summary and Conclusions

Hurricane Florence made landfall in North Carolina in September 2018. Extreme precipitation led to breaches of hog waste lagoons, wastewater facilities, coal ash pits, and other agricultural and industrial sites. After the strong winds and heavy rain had largely passed, extreme freshwater discharge in the Cape Fear River carried various contaminants and debris to the coastal ocean. This contributed to algal blooms, hypoxia, swimming and shellfishing advisories, and other impacts. Here, the ocean pathways of contaminants sourced from the Cape Fear watershed were investigated using the Regional Ocean Modeling System (ROMS) with passive tracers in a river

Onset of water quality hazard after storm landfall: 0–6 days 1–2 weeks 2–3 weeks



Duration of water quality hazard after storm landfall: 0–6 days 1–2 weeks 2–3 weeks or more

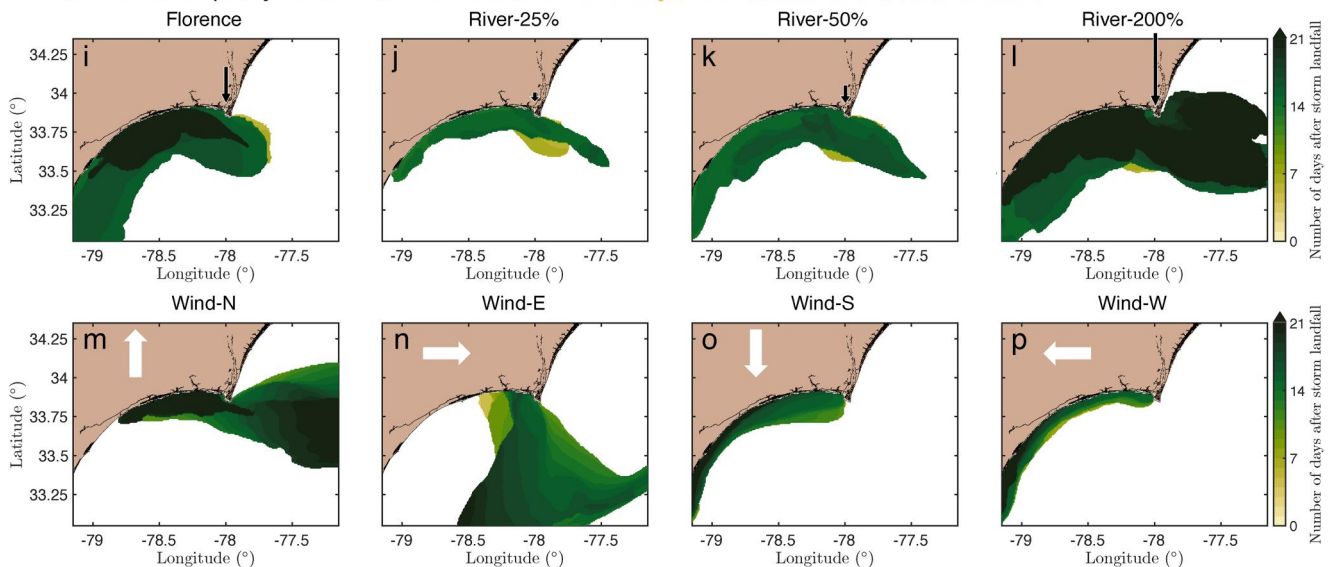


Figure 10. Maps of water quality hazard (a)–(h) onset and (i)–(p) duration (contours of days since storm landfall on 9/14, color scales at right) for land-sourced contaminants in the Cape Fear watershed following Hurricane Florence (a), (i) and a set of scenarios with 25% (b), (j), 50% (c), (k), and 200% (d), (l) the peak river discharge or constant post-storm winds to the North (e, m), East (f, n), South (g, o), and West (h, p) (Table 1) versus Latitude and Longitude. The maps can be used to look up when water quality hazards (here defined concentrations of tracers representing contaminants exceeding a threshold of 5% of the riverine concentration) may begin and end at a particular coastal location of interest following a storm.

point source from a hydrologic-ocean model simulation (ROMS, WRF-Hydro). The time period during and following Hurricane Florence was simulated, along with a suite of synthetic scenarios representing different discharge levels and post-storm wind patterns.

Several days after Hurricane Florence made landfall, when discharge was beginning to rise, contaminants were transported primarily in a westward/downcoast coastal current. Subsequently, during a period with large discharge and alternating upwelling and downwelling winds, material was intermittently stored and released from a bulge/storage region offshore or upcoast of the river mouth. This led to a long tail of elevated tracer concentrations arriving downcoast weeks after the storm. Synthetic extreme events with 200%, 50%, or 25% of the Hurricane Florence discharge have similar spatial patterns, with a more rapid downcoast arrival of contaminants for larger discharge. In a simulation with a weak baseflow discharge level, contaminants remain

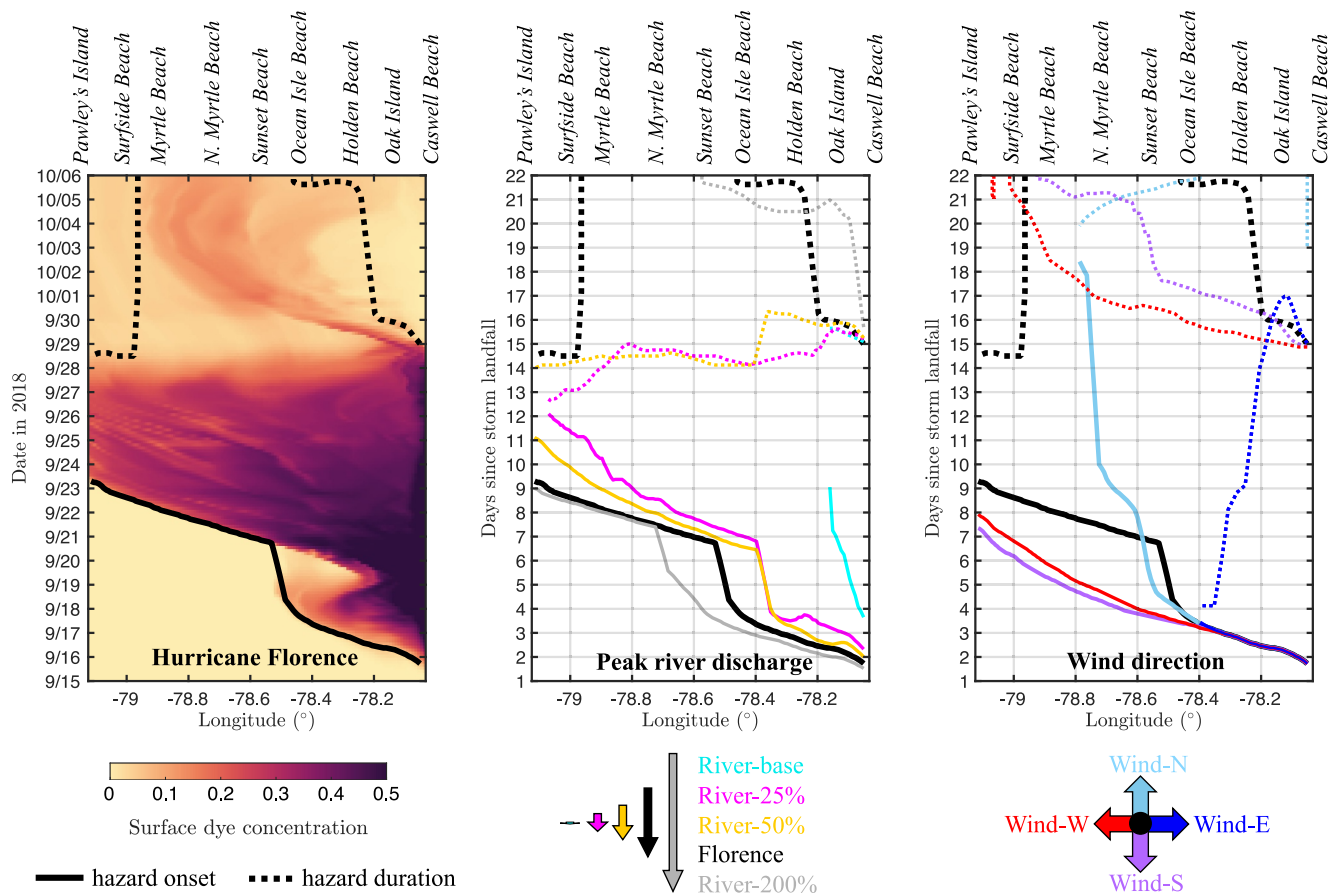


Figure 11. (a) Hovmöller diagram of temporal evolution of surface dye concentration (relative to the riverine concentration, colorbar below) versus Longitude for an alongshore transect (dashed arc in Figure 2c), with lines indicating hazard onset (solid) and duration (dashed), the earliest and latest times when dye exceeds a threshold concentration (b), (c) hazard onset (solid lines) and duration (dashed lines) as a function of time and Longitude along an alongshore transect (dashed arc in Figure 2c) for cases with varying peak river discharge (panel a, legend below) and varying post-storm wind direction (panel b, legend below). Locations of selected beaches are noted above each panel.

confined to a small region near the river mouth. Simulations with fixed wind direction after the storm have a bidirectional pattern for upwelling winds, and an enhanced coastal current for downwelling winds, consistent with the temporal evolution simulated for the Hurricane Florence case. Cases with more downwelling favorable winds had an earlier downcoast arrival of contaminants. Maps and along-coast timestacks of the onset and duration of hypothetical water quality hazards computed for a range of discharge and weather conditions may provide guidance to managers issuing swimming and shellfishing advisories and determining the timing of water quality sampling.

Data Availability Statement

Model configurations and all output needed to reproduce this work are publicly available in a persistent data archive on the Geoscience Data Exchange (GDEX) at <https://doi.org/10.5065/zgm4-sc29> (Moulton et al., 2024).

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Erratum

The originally published version of this article contained errors in Figures 4, 5, 7, and 10. The original figures were cropped to remove the gradients from all but the right-hand panels. The figures have been corrected, and this may be considered the authoritative version of record.